


```
      %¼
[ Guided 6-Face Scan ] %Ä% % % % % % % % % % % % % % %
  (U -> R -> F -> D -> L -> B)      % (Rescan)
      %      %
      %¼      %
[ State Validation Engine ] % % (Invalid) % % (Error Modal Recovery)
      % (Valid)
      %¼
[ Sub-Second Kociemba Solve ]
      %
      %¼
[ Interactive 3D Playback & Step-by-Step Guidance ]
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4. Key UI Components & Interactions

4.1 Guided Camera Scanner HUD (CameraScanner.tsx)

- Viewport Reticle: 3x3 rounded bounding grid centered over the live video stream.
- Stability Indicator: A circular progress ring around each sticker cell filling up over 10 consecutive stable frames.
- Color Confidence Bar: Real-time readout of CIELAB distance matching confidence.
- Face Progress Tracker: 6 segmented indicators at the top showing completed vs remaining faces.
- Orientation Guide Widget: Mini 3D preview cue showing how to rotate the physical cube to reach the next required face.

4.2 Mathematical Error Recovery Modal (ErrorRecoveryModal.tsx)

- Triggered automatically if state validation detects parity or count anomalies.
- Presents an unfolded 2D cube net view (U, L, F, R, B, D).
- Highlights problematic pieces with pulsing red outlines.
- Provides an intuitive color palette selector to manually override misclassified stickers.

4.3 Solve Playback Controls (SolveControls.tsx)

- Move Tape: Horizontal scrolling carousel of move tokens (e.g. [U] [R2] [F'] [D]). The current move is highlighted in bright cyan with enlarged typography.
- Playback Controls:
 - Play / Pause (toggleable with Spacebar).
 - Step Forward / Step Backward (Right Arrow / Left Arrow).
 - Speed Adjustment: 0.5x, 1x, 2x, 4x.
 - Reset to initial scramble (R).
 - Progress Gauge: Real-time percentage indicator and step counter (K / N).

4.4 Real-Time Computer Vision Debugger (CVDebugger.tsx)

- Collapsible diagnostics drawer for advanced users and developers.
 - Displays live RGB, HSV, CIELAB color values for all 9 grid points.
 - Frame rate monitor (FPS) and frame drop detection.
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5. Responsive Design & Accessibility

- Desktop ($\geq 1024\text{px}$): Dual-pane split view with video HUD/controls on the left and the 3D twin visualizer on the right.
- Mobile ($< 1024\text{px}$): Vertically stacked layout with sticky playback controls at the bottom viewport boundary.
- Accessibility:
 - All interactive buttons include descriptive aria-label attributes.
 - High contrast ratio exceeding WCAG AA standards (minimum 4.5:1 for normal text, 7:1 for large display metrics).
 - Keyboard navigation fully supported across all critical playback actions.